

// GLOBEGRID v6 -- BUILD MANUAL
// GROQ-FIRST LLM ARCHITECTURE -- SEE SECTION 01

GLOBEGRID v6

WAR MODE, WORLD DATA DEPTH, AND ACCURACY

Dense format. Every item is a direct build instruction.

REVISION	6.0
BUILDS ON	v1-v5
LLM PROVIDER	Groq is now PRIMARY for all LLM calls (Analyst, translation, synthesis) -- not a fallback. See Section 01.

SECTION 01 // LLM PROVIDER

Groq-first architecture (critical, do this first)

Overrides v5 s18's fallback order. Groq is now PRIMARY, not a fallback, specifically because Analyst answers and translation (this manual, Sections 11 and 29) are high-volume, latency-sensitive call sites where Claude API credit cost was the explicit problem.

- `llm_provider.primary = "groq"`, model: Llama 3.3 70B via Groq's OpenAI-compatible endpoint (free tier: ~14,400 req/day, no card).
- Applies to: Analyst panel answers (v3 s24), translation (Section 11 below), title/grammar correction (Section 3), country/entity synthesis (v3 s13.2), web-search-grounded fact verification (Section 30).
- Causal-link narratives (v1 s9) MAY stay on Claude if quality there matters more than cost at low volume -- keep this one call site independently configurable, everything else defaults to Groq.
- Provider abstraction (v5 s18) already supports swapping via config with no call-site changes -- this is a config flip plus verifying every LLM call site actually reads from the shared provider config rather than hardcoding a client.
- Fallback order if Groq rate-limits: Gemini free tier, then Ollama local, then Claude (cost fallback of last resort, not first resort).

SECTION 02 // REMOVE GDELT

Remove GDELT Events and GDELT Cloud entirely

Explicit instruction, no argument needed: remove both GDELT sources (v1 s4.2) completely. Delete the ingestion modules, remove from sources seed data, stop polling.

- Replacement coverage comes from Section 7 below (broader reputable outlets, gov/mil comms, curated social) -- do not leave a coverage gap, this removal and the source expansion should ship together.
- Any existing ingested GDELT-sourced rows in the DB can stay (historical fact chain is never deleted, v1 s5.9) but the source itself stops polling immediately.

SECTION 03 // CONTENT FORMAT

Bulleted synthesis + standardized titles

- Story synthesis (v1 s5.5) renders as digestible bullet points by default, with a 'full summary' expand button revealing the fuller prose synthesis (v4 s11.2) -- two densities of the same underlying content, not two different generations.
- Title standardization rules (extends v4 s11.1): consistent capitalization (sentence case), no emoji unless factually part of the story (rare), no unfinished sentences, hard length cap with clean truncation, no raw links (v5 s1).
- Grammar/spelling correction (v4 s11.1) explicitly applies across every supported language (Section 11), not just English -- runs through the Groq-first pipeline (Section 1).

SECTION 04 // MILITARY CATEGORY**Confirms v5 s3 -- no new spec needed**

Already specified: events.development_type = 'military' covers tech developments, tests, drills, movements outside active war (v5 s3). Reaffirmed here per the same request arriving again -- verify it actually shipped, don't re-spec it.

SECTION 05 // LEADER & PARTY PROFILES

Rich wiki pages for leaders and parties

Extend: political_parties (v4 s6.2)

Field	Type	Description
ideology_tags	text[]	Already exists -- confirm populated, not just schema
electoral_history	text (JSON)	Past election results this party contested
coalition_partners	text[]	Current governing/opposition coalition, if any
logo_image_url	text	Wikimedia-sourced, same CC pattern as flags (v5 s7)
founding_history	text	AI-synthesized, source-linked

Extend: notable_persons (v4 s22)

Field	Type	Description
electoral_history	text (JSON)	Elections this person won/lost/contested
portrait_image_url	text	Wikimedia-sourced

Wiki-style prev/next navigation and fullscreen toggle (v4 s6.2) already apply -- these are new fields on existing entity types, not a new page system.

SECTION 06 // RECTANGLE SELECT

Drag-to-select events by area

- Drag a rectangle on globe/map; on release, list every event inside it, grouped by location, in the left sliding pane (v4 s17-18) -- reuses the same list-content template built for the cluster-overflow fix (v5 s9).
- **The rectangle is always screen-axis-aligned** (upright, parallel to window edges) regardless of globe rotation or map projection -- draw it in screen space, not world space, like a PDX-game selection box. Never rotate or skew it.

SECTION 07 // BORDERS & BLOCS

Border, bloc, and territory data fixes

Fixes

Item	Fix
Ukraine occupied regions	Add all four Russian-occupied Ukrainian regions to border_disputes (v4 s5.3) as ONE dispute entry (de-facto annexation, still internationally disputed) -- claimant_a = UKR, claimant_b = RUS, single territory_name covering all four, status = active.
Border/coastline mismatch	Country border resolution and continent outline resolution must match (both use the 10m dataset from v4 s2.3) -- currently borders render smoother/lower-detail than coastlines, causing visible seams. Single consistent resolution for both layers.
Multi-bloc toggle	Alliance overlay (v3 s14, v4 s4.1) becomes multi-select: toggle any combination on simultaneously, an 'all' shortcut, and category shortcuts (e.g. 'military blocs only').
More blocs (SCO, etc.)	Expand alliance seed data (v3 s14, synced via v4 s5.1's Wikidata approach) to include regional cooperation orgs like SCO and similar -- inclusion bar is 'functions as a real geopolitical force,' not 'majorly covered by Western media.'
Timor-Leste in ASEAN	Add directly -- Timor-Leste is ASEAN's 11th member. Same root-cause class as v4 s1.1's other missing-membership bugs: verify the sync (v4 s5.1), don't just patch this one row.

SECTION 08 // WAR MODE

Dedicated conflict layout mode

Biggest new feature in this manual. A distinct view entered from a conflict's page/tab (Section 9) or an Analyst deep-link (Section 13).

- **Map:** zoomed/framed to the conflict zone; all parties on one side rendered in one consistent color per side (uses `conflict_parties`, v3 s15); minor/sub-national factions and detailed internal borders that never show in global mode appear here (e.g. Tuareg-controlled areas within Mali) -- these need their own lightweight sub-territory data, scoped to conflicts, not the global countries table.
- **Left panel:** temporarily replaces the feed (not permanently -- pop-ups/pane behavior otherwise unchanged) with a wiki-infobox-style overview: belligerents/sides, key facts, dates, status -- same visual language as the country/wiki panel system (v4 s6), a new content template on the same shell.
- **Right panel:** event feed restricted to this conflict only, sub-filtered into Military / Civilian / Diplomatic / Economic tabs (reuses `development_type`, Section 4, plus a broader per-event category tag where those four values apply).

New table: `conflict_subfactions`

Field	Type	Description
<code>id</code>	<code>uuid</code> PK	
<code>conflict_id</code>	<code>uuid</code> FK	References <code>conflicts.id</code>
<code>name</code>	<code>text</code>	e.g. 'Azawad / Tuareg-controlled area'
<code>zone_geojson</code>	<code>text</code>	Rough area, same style as non-state-actor zones (Section 21)
<code>side</code>	<code>text</code>	Which side of the conflict this faction aligns with, if any

SECTION 09 // CONFLICTS TAB**Top-level Conflicts tab; remove redundant listing**

- Add 'Conflicts' as its own top-header tab, next to Stories -- conflict tabs (v3 s15) reachable directly, not only discovered from within a story.
- Remove the existing redundant 'All' conflicts listing above the right-panel event feed -- that functionality now lives in the new Conflicts tab.
- Selecting a conflict here enters War Mode (Section 8) directly.

SECTION 10 // FAST MULTILINGUAL INGESTION**Faster pulling + immediate translation at ingestion**

- Add faster-polling, reputable non-English sources per conflict relevance (e.g. Ukrainian/Russian outlets for that war) so local-language reporting is ingested before Western wire coverage catches up -- extends v5 s5's source expansion and v2's non-English outlet list.
- Translation happens at ingestion time for the purpose of correlation/synthesis (feeding v3 s4's cross-lingual embeddings) -- this is separate from display-time translation (Section 11), which is about what the user sees.
- Runs on the Groq-first pipeline (Section 1) -- this is high-volume, exactly the case Groq's free tier is meant for.

SECTION 11 // LANGUAGE BUTTON

Site-wide instant translation

- Header 'Language' button (also duplicated outside Settings per explicit request) sets a site-wide language preference -- every panel, label, and piece of content translates immediately, including content that arrives after the setting is changed (new stories, live feed items) -- not a per-item 'translate' button anymore.
- Implementation: cache translated strings per (content_id, language) pair so re-rendering is instant after the first translation; new content translates on arrival via the same Groq-first pipeline (Section 1), not on first view.
- Non-Latin-script languages: render a phonetically accurate transliteration of 'GlobeGrid' next to the Latin wordmark in that script (e.g. Devanagari, Cyrillic, Armenian) -- a small, fixed lookup table per supported language (v5 s2's language list), not machine-transliterated on the fly (brand names need a human-reviewed, consistent spelling, not a fresh guess each time).

SECTION 12 // ESC KEY BEHAVIOR

Escape closes panels one at a time

- Esc closes the topmost/rightmost open panel first; a second Esc closes the next one; and so on -- a real panel stack, not a single global close-all.
- Consistent with the shared pane system (v4 s18) -- implement as one keydown handler querying the stack, not per-panel listeners.

SECTION 13 // NAVIGATION FIXES

Deep links, camera sync, no random panning

Fixes

Item	Fix
Analyst deep link to War Mode	When an Analyst answer opens a conflict (v3 s24.4), it must enter War Mode (Section 8) directly and frame/zoom to the conflict zone immediately -- not just open a generic conflict tab.
Feed click camera sync	Clicking an event in the live feed list must move the globe/map camera to that event's location -- currently doesn't.
Random panning bug	Map/globe should never move on its own outside of the idle-tour feature (v2 s5.1), which itself only fires after real inactivity and is toggle-off-able. Audit for any stray auto-pan/reset call firing unexpectedly and remove it.

SECTION 14 // TERRITORY PAGES

Greenland, Puerto Rico, and similar

- Add `countries.status = 'territory'` (extends v4 s5.2's status enum) for major territories -- Greenland, Puerto Rico, and similar -- each gets its own profile page (population, stats, etc. per Section 15) plus a visible link to its sovereign/'overlord' country's page (new `countries.sovereign_id`, self-referencing FK).
- Currently clicking these goes nowhere -- this is a real gap in the entity-completeness fix (v4 s5.1), not just a display issue: they need to exist as clickable entities in the first place.

SECTION 15 // COUNTRY PROFILE DEPTH

More stats, official names, guaranteed completeness

Extend: countries

Field	Type	Description
official_name	text	e.g. 'Islamic Republic of Afghanistan' -- displayed under the common name in smaller font
population	integer	Synced alongside leadership data (v3 s13.2 cadence)
languages	text[]	
gdp_usd	numeric	Already partially covered by v3 s13's country_trade_stats -- consolidate or cross-reference, don't duplicate
hdi	numeric	Human Development Index
gdp_per_capita_usd	numeric	

- Past conflicts list added to each country's conflicts section (alongside current ones, v3 s13) -- query conflict_parties (v3 s15) for status = resolved, not just active.
- **Zero-empty-entry guarantee:** every country must have a complete profile immediately -- no missing leader portraits, no blank fields. This directly fixes the Afghanistan case: reclassify the Taliban government as Afghanistan's de_facto leadership (status field, v4 s5.2) rather than leaving the country entry empty (also see Section 21's related non-state-actor reclassification).
- Sourced from government sites, Wikidata, and Wikipedia per the existing background-sourcing pattern (v4 s7) -- extend that pipeline's completeness check (v4 s5.1) to fail loudly on any country with empty required fields, not just wrong membership counts.

SECTION 16 // THEMATIC MAP MODES

CIA-Factbook-style interactive data maps

Vision alignment: this is the feature that makes GlobeGrid a genuine successor to the CIA World Factbook, not just a news dashboard -- treat data accuracy here as non-negotiable, not best-effort.

- EU5-style UI: a button that rolls out a hoverable icon bar (matches the aesthetic direction already established) offering map modes: HDI, GDP, GDP per capita, population, population density, dominant religion (country-level AND sub-national/area-level as distinct modes), dominant language (country-level and area-level), dominant religious denomination by region.
- **Zero tolerance for inaccuracy** on demographic data specifically -- source from authoritative datasets only: World Bank Open Data (GDP, population, HDI proxy via UNDP), UN Statistics Division, Pew Research (religion), Ethnologue or UN language data (language). Never LLM-guessed demographic figures.
- Choropleth rendering: color country/region polygons by the selected metric's value, using the same boundary data as the base map (Section 7) so shapes stay consistent across modes.
- Add more modes as natural extensions once the pattern is built (e.g. literacy rate, urbanization rate, median age) -- the button/icon-bar UI should be built to accept new modes cheaply, not hardcoded to a fixed set.

SECTION 17 // AUDIO

Fix Ambient's buzz bug; test ONE new preset only

Explicit scope narrowing from the previous ask: do NOT build all remaining presets from v4/v5. Fix the Ambient bug, then build and test only Ambient Hard Rock/Metal as a proof of concept before doing anything further.

- **Ambient buzz bug:** a constant background buzzing suggests an unintended DC offset, an unclosed/leaking oscillator node, or a feedback loop in the drone's gain envelope -- audit the Ambient Default signal chain specifically for a node that's never stopped/disconnected between plays.
- **Metal/hard rock test:** build using the real primitives already specified (v5 s17: WaveShaperNode for distortion, BiquadFilterNode for tone shaping, proper ADSR) -- the explicit ask is 'I want to hear the guitars,' meaning real distortion character, not a clean oscillator with a metal label on it.
- Confirm this one preset sounds right before touching vaporwave, Dune, technohouse, Berlin industrial, war-room orchestral, or numbers station -- those stay unbuilt until this proof-of-concept is validated.

SECTION 18 // 2D MAP PERFORMANCE

Still laggy -- deeper fix needed

v5 s12's viewport-culling fix was insufficient (still laggy, if less than before). Likely cause: the 3-tile wraparound approach (v4 s4.2) still renders full tile geometry eagerly even with culling applied per-tile.

- Move to a real tile-based rendering scheme (small fixed-size tiles loaded/unloaded based on viewport, standard web-map architecture) instead of three full-map copies -- this is the actual fix, not a tuning pass on the existing approach.
- Confirm with the browser's performance profiler before and after -- don't ship this fix on 'feels better,' verify frame time and memory directly given this is now the second attempt.

SECTION 19 // TERRAIN TEXTURE FIX**Real biome/elevation textures, proper occlusion**

- Visible-through-the-globe bug is the same root-cause family as v4 s2.1 (missing proper depth/occlusion handling) -- apply the same camera-facing fix, this time to the terrain shading pass specifically.
- Replace whatever placeholder terrain texture exists with a real biome/elevation-driven one: NASA Blue Marble (biome color) blended with SRTM-derived elevation shading (v5 s8's original source suggestion) -- properly sampled and mapped to the globe's UV space, not a generic noise texture.
- Toggle (v5 s8) and LOD-gating (v5 s8, v4 s2.3) stay as specified -- this section is strictly about rendering correctness and texture quality, not the toggle mechanism.

SECTION 20 // PORTRAIT/FLAG SIZING**Fix leader portrait vs flag proportions**

Country panel header (v4 s6.1, v5 s7): normalize both images to the same vertical height (not the same width -- flags and portraits have different natural aspect ratios) so neither looks disproportionately tiny next to the other.

SECTION 21 // NSA ZONES

Real zone shapes; Taliban/Afghanistan reclassification

- Replace rectangular zone rendering (v5 s11) with actual shaped polygons rendered as a pulsing or dotted/shaded rough-boundary style -- not solid rectangles. Visual treatment, not a schema change (zone_geojson already supports real polygons; the rectangles were a rendering shortcut, not a data limitation).
- Clicking a zone opens that non-state actor's overview page (v3 s16) directly -- currently not wired up.
- **Taliban/Afghanistan fix:** the Taliban is Afghanistan's governing authority as of 2026, not a non-state actor. Reclassify: remove from non_state_actors, set Afghanistan's `countries.status = 'de_facto'` (v4 s5.2) with the Taliban leadership populated as Afghanistan's head_of_government (v3 s13.2) -- this also directly resolves the 'empty Afghanistan entry' problem from Section 15.
- This is a completeness-check case worth generalizing: any non-state actor that has become a country's actual governing authority should trigger a similar reclassification review, not just this one instance.

SECTION 22 // INSTABILITY RECALIBRATION**Fix unrealistic scores**

- Current formula (v1 s5.6) reads too high at normal baseline (e.g. 73 on a routine day). Recalibrate weights/normalization against a rough historical baseline: routine day should sit low (e.g. 10-25 range), only genuinely severe multi-region escalation should approach the top of the scale.
- Practical approach: run the existing formula against several weeks of real ingested data once ingestion volume is up (post Section 2/7 source changes), then adjust the weight constants (v1 s7.2 config) empirically rather than guessing again -- this is a tuning pass that needs real data, not a formula rewrite.

SECTION 23 // MORE THEMES

10 additional themes

Extends v5 s14's theme list. Add 10 more, including at least: a black-red high-contrast 'edgy' theme, plus others in a similar spirit (e.g. monochrome-blue 'cold war', deep-purple 'nightfall', high-saturation 'neon'). CSS-variable-swap implementation (v5 s14) already supports this cheaply -- this section is content (the palettes), not new mechanism.

SECTION 24 // CITY LIGHTS TOGGLE**On/off setting**

Simple settings toggle for the existing night-side city-light effect (v1 s7's CITY_LIGHTS usage) -- currently always on, no way to disable.

SECTION 25 // LOCATION PRECISION

Fix wrong-city geocoding

Reported case: an article about Khamenei's funeral in Tehran incorrectly geocoded elsewhere (e.g. Delhi). This is a geocode-confidence issue (v4 s3) at the extraction level -- the pipeline is picking up an incidental place mention instead of the article's actual dateline/primary location.

- Fix: prioritize the article's dateline or lead-paragraph location over any place name mentioned elsewhere in the body text -- a name like 'Khamenei' or a country mentioned in passing should not outrank an explicit dateline.
- Where no dateline exists, weight the FIRST explicit place mention in the text over later ones, rather than any-match-wins.

SECTION 26 // PULSING HIGHLIGHT

Animated border highlight on focus

When a country/region (or multiple countries, e.g. from a region-level Analyst query, v5 s20) is opened via profile or Analyst, outline it on the globe/map with an animated pulsing current-style border -- reuses the shared animation system (v4 s18) for the pulse, and the highlight clears when the corresponding panel closes.

SECTION 27 // STORY THREADS

Faster directory; macro-trend grouping

Two related items: performance, and a real content-modeling gap.

- **Performance:** profile the Stories directory (v4 s8.2) load path -- likely missing pagination or an index on the query filtering by story_type/region; fix directly rather than guessing.
- **Story Threads (new):** individual stories (v1 s5.4 clusters) are too granular for macro trends like 'Strait of Hormuz Developments' or 'EU-Russia Tensions.' Add a new grouping layer above individual stories, not a replacement for them.

New table: story_threads

Field	Type	Description
id	uuid PK	
name	text	e.g. 'Strait of Hormuz Developments'
description	text	AI-synthesized overview of the ongoing trend
first_seen_at	timestampz	
last_updated_at	timestampz	

New table: story_thread_members

Description		
	uuid FK	References story_threads.id
	uuid FK	References stories.id

Threads are the primary browsing unit in the Stories directory going forward; individual acute_event stories remain visible within a thread, not hidden -- this reorganizes presentation, it doesn't remove the underlying story_type taxonomy from v4 s8.1.

SECTION 28 // REGION COMPLETENESS

Fix incomplete region groupings

Confirmed-pattern bug, same family as v4's missing-country/alliance issues: South Asia is missing Nepal, Bhutan, Sri Lanka, Maldives; Eastern Europe has only 8 countries. countries.region values were hand-assigned rather than synced against a standard taxonomy.

- Fix: assign countries.region from a standard geographic classification (UN M49 region/sub-region codes are a reasonable authoritative source) synced the same way as the Wikidata-driven completeness fix (v4 s5.1), not hand-picked lists.
- Add this to the same startup completeness check (v4 s5.1) -- verify every UN M49 sub-region's expected country count matches what's actually assigned, the same pattern already used for total country count.

SECTION 29 // ANALYST QUALITY BAR

The linchpin: what 'actually smart' means here

Synthesizes every Analyst-related request in this manual and prior ones into one concrete bar.

- **Screen-aware:** reads whatever panel/page is currently open (extends the context-aware focus in v3 s24.1, correctly scoped this time per the v4 s16 fix) and factors it into answers without needing it restated.
- **Region queries produce real deep-dive pages:** a region-level answer (v5 s20) must link the relevant countries AND relevant Stories/Story Threads (Section 27) AND individual Events -- not just countries. 'Eastern Europe' should surface Ukraine War developments, NATO expansion, border armament as actual linked content, not a one-line summary.
- **Bulleted, with a deep-dive option** -- same content-density pattern as Section 3's story synthesis, applied to Analyst answers.
- **Runs on Groq first** (Section 1) -- speed matters for this to feel responsive given how central it is to the product.
- **Grounded in Section 30's web-search verification** where the question concerns a fact that might have changed since the last sync, rather than answering from potentially stale cached synthesis alone.

SECTION 30 // WEB-SEARCH ACCURACY PIPELINE

Keep everything correct via live search

The Khamenei-era bug (v5 s15) and every 'missing/outdated data' item in this manual point at the same root fix: give the refresh pipeline actual live web search, not just scheduled structured-source syncs.

- Add a free/cheap search API call (e.g. Brave Search API free tier) ahead of each scheduled refresh job (leadership sync, alliance membership sync, country stats, v3 s13.2 / v4 s5.1) -- fetch current top results for a targeted query ('current head of state of X'), then have the Groq-first pipeline (Section 1) synthesize/verify against those results before writing the refreshed value.
- This keeps the 'search the web to fix inaccuracies' request inside the Groq-first, low-cost architecture -- the search API supplies fresh facts, Groq does the cheap synthesis step, no dependency on a provider with a built-in search tool.
- Also run this pipeline on-demand when the Analyst (Section 29) detects its cached synthesis might be stale for a question asked live, not only on the scheduled cadence.

SECTION 31 // HIDE VR BUTTON

Non-functional settings item

WebXR (v3 s9) isn't functional yet -- hide the settings button until it actually works, rather than shipping a dead control. Don't remove the underlying spec, just the visible entry point.

SECTION 32 // CONFIG ADDITIONS**New config.yaml values**

```
llm_provider:
  primary: "groq"                # was "anthropic" -- see Section 01
  groq_model: "llama-3.3-70b-versatile"
  fallback_order: ["gemini_free", "ollama_local", "anthropic"]
  causal_link_override: "anthropic" # optional, quality-critical path only

sources:
  gdelt_events_enabled: false      # removed, Section 02
  gdelt_cloud_enabled: false       # removed, Section 02

translation:
  instant_translate_on_arrival: true
  cache_by_content_and_language: true

war_mode:
  enabled: true

audio:
  presets_active: ["ambient_default", "arctic_calm", "modal_drift", "data_sonification", "hard_rock_metal"]
  # everything else from v4/v5 stays unbuilt until hard_rock_metal is validated

map_rendering:
  tile_based_2d: true             # replaces the 3-tile wraparound approach

instability:
  recalibration_pending: true      # tune after real ingestion volume from Section 02/07 changes

accuracy:
  web_search_verification_enabled: true
  search_provider: "brave_free_tier"

ui:
  vr_button_visible: false
```

SECTION 33 // BUILD SEQUENCE

Build order

Phases

Item	Fix
Phase A6	LLM provider switch to Groq (S1) + remove GDELT (S2) -- both are config/removal, do first, unblocks cost concerns immediately.
Phase B6	Taliban/Afghanistan reclassification (S21/S15) + region completeness (S28) + border/bloc data fixes (S7) -- all data-correctness fixes, batch as one data pass.
Phase C6	War Mode (S8) + Conflicts tab (S9) -- the flagship new feature, build together.
Phase D6	Language button + instant translation (S11) + fast multilingual ingestion (S10) -- both language-pipeline work, do together, on Groq.
Phase E6	Navigation fixes (S13) + ESC stack (S12) + rectangle select (S6) + pulsing highlight (S26) -- all interaction-layer polish, one pass.
Phase F6	Country profile depth (S15) + territory pages (S14) + leader/party profiles (S5) -- all entity-content depth, one pass.
Phase G6	Thematic map modes (S16) -- large, standalone, source data carefully per its zero-tolerance-accuracy requirement.
Phase H6	Story Threads (S27) + directory performance -- new grouping layer, do after War Mode since both touch story presentation.
Phase I6	2D map tile-based rewrite (S18) + terrain texture fix (S19) -- both rendering-correctness, do together, profile before/after.
Phase J6	Web-search accuracy pipeline (S30) -- depends on Phase A6's Groq setup; wire into existing refresh jobs.
Phase K6	Audio: fix Ambient buzz + build/validate hard rock/metal only (S17) -- explicitly nothing further.
Phase L6	Everything else (S3, S4 confirm, S20, S22-25, S31) -- small, independent, fold in wherever convenient.

// END OF MANUAL -- GLOBEGRID v6 BUILD MANUAL -- REV 6.0